

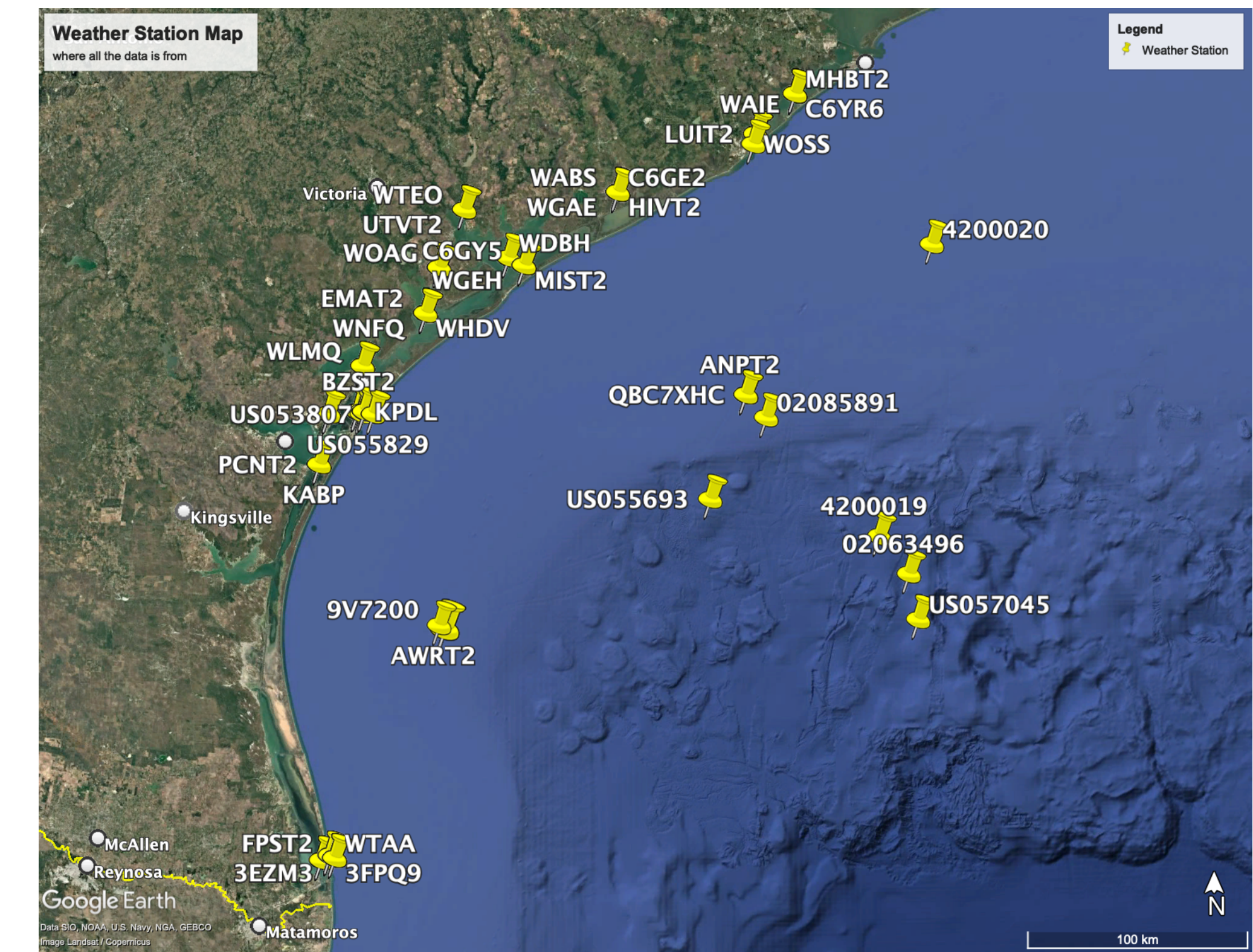
Mean Gulf Temperatures in May-Region 1

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Have air temperatures increased in the gulf over the last 100 years; is the climate really changing?

Introduction: Using Julia Statistical Analysis and Loess (locally estimated scatterplot smoothing), I attempt to analyze weather station data in gulf region 1 over the last 100 years, finding true changes in climate with available data. Attempting to analyze beginner trends in changing of the climate attributed to rising local temperatures from antropogenic pollution. I hypothesize that air temperatures have slowly risen over the last hundred years.

- Data from weather stations within coordinate grid (29.25, -97.25 to 25, -94.5), showing 100 year air temperature data from the inner gulf coast
- Julia (pkgs: CSV, DataFrames, Dates, Plots, RollingFunctions, Statistics, Loess), Google Earth Pro,
- This code snippet turns the mean august temperature each year into an overall trend by fitting with a Loess Curve
- This is after, of course, turning the entire data set that contained every weather station reading into a data set that took the average temperature of each year.



Map of the Weather Stations within coordinate grid.

The data below show that mean air temperature has a rising trend over the last 100 years. An interpolated trend of the Loess curve shows a positive slope, indicating a rise over time, though the sinusoidality of the Loess curve is apparent.

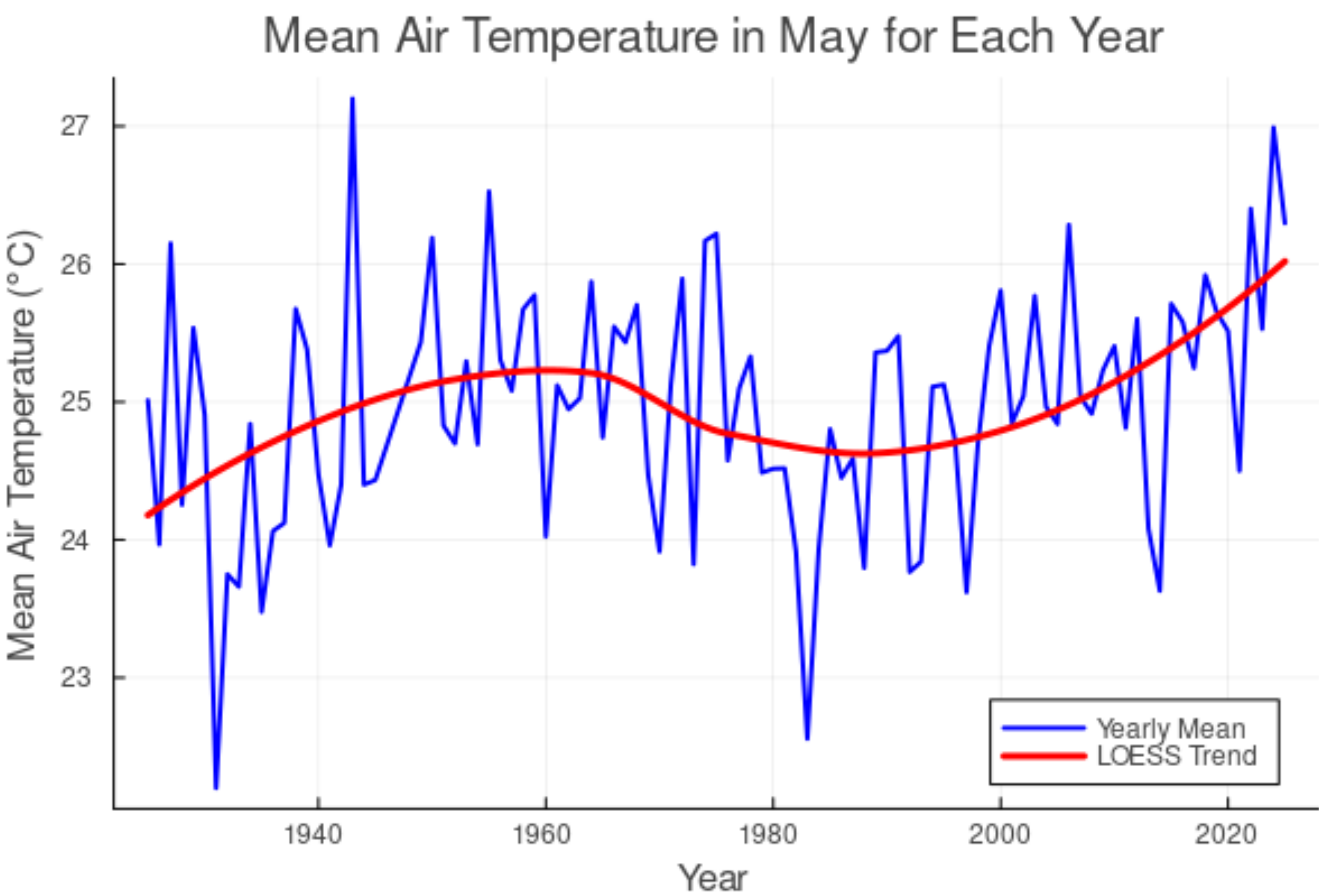


Figure of the mean air temperature in May for 1925-2025.

While the LOESS curve might not be the best representation of trend, it is accurate without removing too much noise. The sinusoidality of the curve shows a cyclical nature, but there is still an upward trend of the data, signalling that even though the temperature might go down, over a 50 year climate cycle, the pattern is increasing. This may signal that even though temperatures may decrease a bit in the future relative to the last 20 years, relative to the last one hundred they are steadily increasing. One might attribute this increase to an increase of greenhouse gas buildup in the atmosphere.

In short, the overall average temperature of the gulf region is increasing by a significant extent. The slow increase could help to explain to readers why we havent "noticed any effects yet", since climate is an average over many years and any change would happen spread over many years as well. The average temperatures in May in Gulf Region 1 have most definitely increased, and in the future, have no statistical reason not to continue to do so.

This is evident by Julia dataset analysis from the recordings taken across a multitude of sites on the inner gulf coast.